AN ELEMENT THAT IS IDEMPOTENT MODULO AN IDEAL SATISFIES A MONIC POLYNOMIAL THAT FACTORS AS X-D(X-1)-D MODULO THE IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- Integrality criterion over an ideal: an element of an integral extension lying in the extended ideal satisfies a monic polynomial whose remaining coefficients lie in the ideal. the author cites Algebra, Lemma algebra-lemma-integral-integral-over-ideal at proof lines 1968-1971. Supplies the auxiliary monic polynomial g whose coefficients below the top one lie in the ideal.
- Behaviour of polynomial composition with the quadratic x squared minus x modulo the ideal. used at proof lines 1971-1974. Produces the announced shape of the reduced polynomial.

Source reference. Original lemma 09XG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12529–12529. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00H5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 631–631. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let I be an ideal of a ring A. Let $A \to B$ be an integral ring map. Let $b \in B$ map to an idempotent in B/IB. Then there exists a monic $f \in A[x]$ with $f(b) = 0$ and $f \bmod I = x^d(x-1)^d$ for some $d \geq 1$.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Observe that $z = b^2 - b$ is an element of IB. By Algebra, Lemma 00H5 there exist a monic polynomial $g(x) = x^d + \sum a_j x^j$ of degree d with $a_j \in I$ such that $g(z) = 0$ in B. Hence $f(x) = g(x^2 - x) \in A[x]$ is a monic polynomial such that $f(x) \equiv x^d(x-1)^d \bmod I$ and such that $f(b) = 0$ in B. $\square$